

Software Requirements Specifications

Alzheimer's Aid System

Project Code:

Internal Advisor:

Professor Fahad Maqbool

Project Manager:

Professor Dr Muhammad Ilyas

Project Team:

Isha Tariq (BSCS51F21S063)
Fatima Waseem (BSCS51F21S090)
Areeba Imtiaz (BSCS51F21S072)

Submission Date:

26-November-2024

Project Manager's Signature

Document Information

Customer	University of Sargodha
Project	Alzheimer's Aid System
Document	Requirement Specifications
Document Version	1.0
Identifier	
Status	Draft
Author(s)	Isha Tariq, Fatima Waseem, Areeba Imtiaz
Approver(s)	PM
Issue Date	Nov. 26, 2024
Document Location	
Distribution	1. Advisor 2. PM 3. Project Office

Definition of Terms, Acronyms and Abbreviations

Term	Description
ML	Machine Learning
AI	Artificial Intelligence
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
MRI	Magnetic Resonance Imaging
GPS	Global Positioning System
RS	Requirements Specifications

Table of Contents

SOFTWARE REQUIREMENTS SPECIFICATIONS	1
1.INTRODUCTION	4
2.OVERALL SYSTEM DESCRIPTION.....	4
3.EXTERNAL INTERFACE REQUIREMENTS	6
4.FUNCTIONAL REQUIREMENTS	6
5.NON-FUNCTIONAL REQUIREMENTS	9
6.ASSUMPTIONS AND DEPENDENCIES	10
7.REFERENCES.....	11
8.APPENDICES.....	11
SUPPORTING DETAILS	11
8.1 Datasets	11
8.2 Security Standards	11
8.3 Tools and Technologies	12
8.4 User Interfaces	12
8.5 Testing Plans	12
8.6 Future Extensions.....	12

1. Introduction

1.1 Purpose of Document

This document acts as the Software Requirements Specification (SRS) for the Alzheimer's Aid System. It details the project's requirements and design intended for stakeholders, developers, and researchers. The main goal is to offer a thorough understanding of the system's features, functions, and constraints.

1.2 Project Overview

The Alzheimer's Aid System is a platform powered by AI and machine learning, aimed at improving the support provided to individuals with Alzheimer's. It offers tools for:

- Identifying early signs of cognitive decline through advance AI models.
- Providing memory support, such as reminders for daily activities and medications.
- Offering an interactive voice-activated assistant to boost user interaction.
- Enabling remote monitoring of patients for caregivers, which includes GPS tracking and notifications.
- By utilizing state-of-the-art technologies, this initiative seeks to enhance patients' quality of life and alleviate the strain on caregivers.

1.3 Scope

The Alzheimer's Aid System will comprise:

- Modules for early detection utilizing graph-based AI technologies.
- Customized cognitive support and memory aid tools.
- A voice-controlled assistant to help with daily activities.
- GPS tracking in real time, along with alerts for caregivers.

The system will exclude:

- Telemedicine functionalities.
- Support for conditions other than Alzheimer's in its initial release.

2. Overall System Description

2.1 User characteristics

The main users include:

Patients: Older adults experiencing different degrees of cognitive decline.

Caregivers: Relatives or healthcare workers who oversee the patients.

2.2 Operating environment

The system will function on:

Devices: Smartphones, tablets, and computers.

Platforms: Web applications that support Windows.

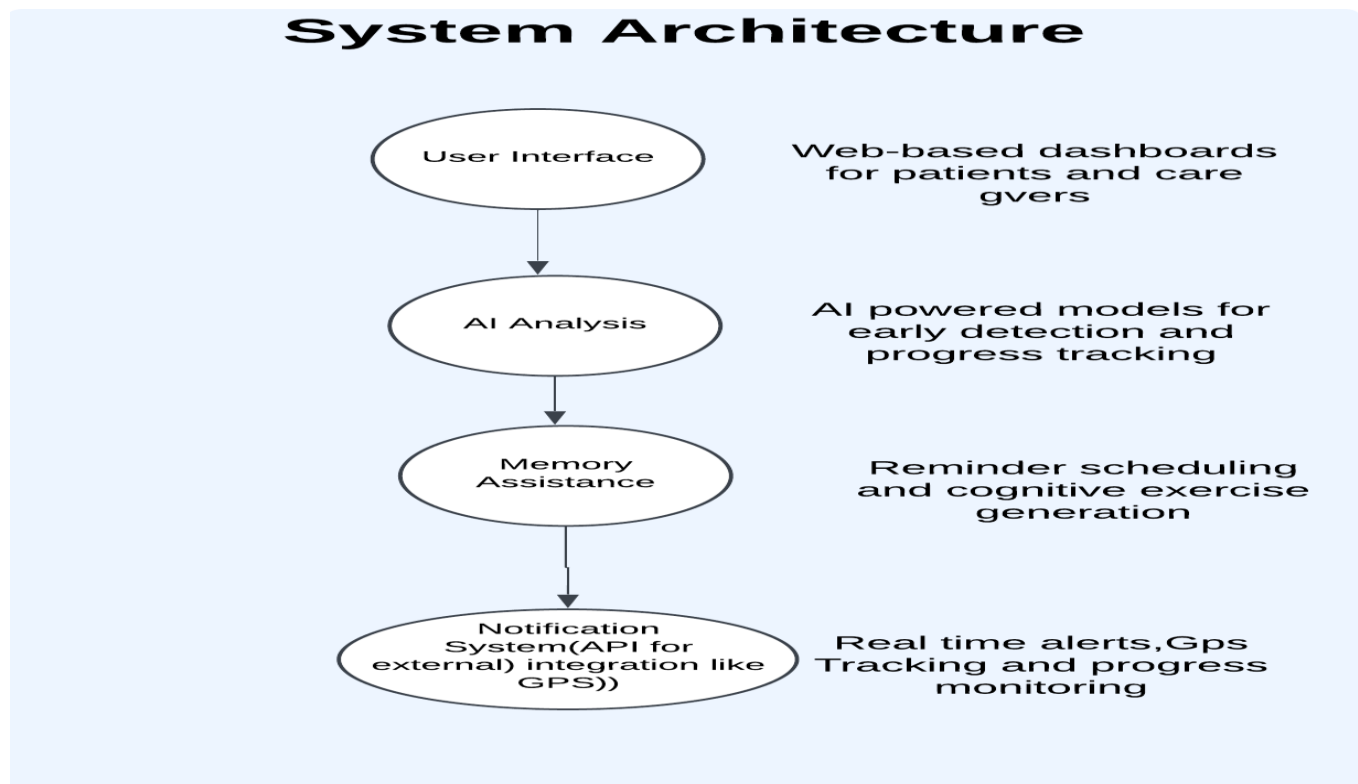
Requirements: A stable internet connection is necessary for data synchronization and real-time functionalities.

2.3 System constraints

The system must adhere to these constraints:

Data Protection Laws: Compliance with HIPAA and various other privacy standards is required.

Device Constraints: Compatibility must be ensured with devices that have limited resources.



3. External Interface Requirements

3.1 Hardware Interfaces

- Devices equipped with GPS for tracking patients.
- Devices featuring microphones for voice communication.

3.2 Software Interfaces

- Compatibility with graph databases (such as Neo4j).
- Handling clinical and imaging data sourced from ADNI and KAGGLE datasets.
- Connectivity with external services (API) for anomaly detection and automated alerting

3.3 Communications Interfaces

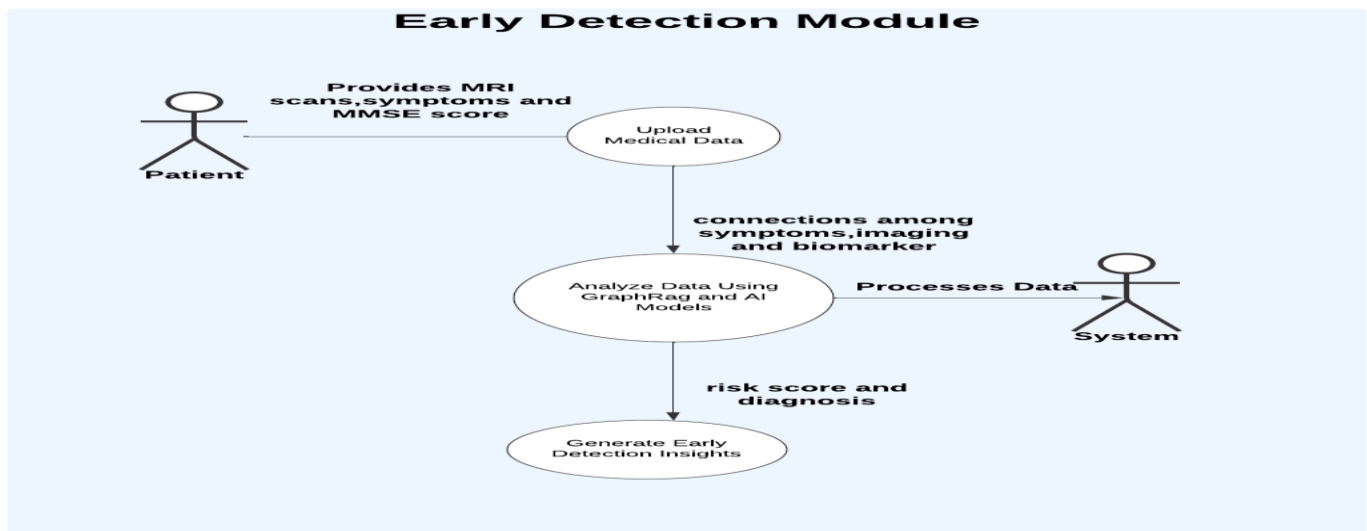
- Secure data transmission through HTTPS protocols.
- Alerts sent via email and mobile push notifications.

4. Functional Requirements

Following are the functional requirements along with their use case diagrams:

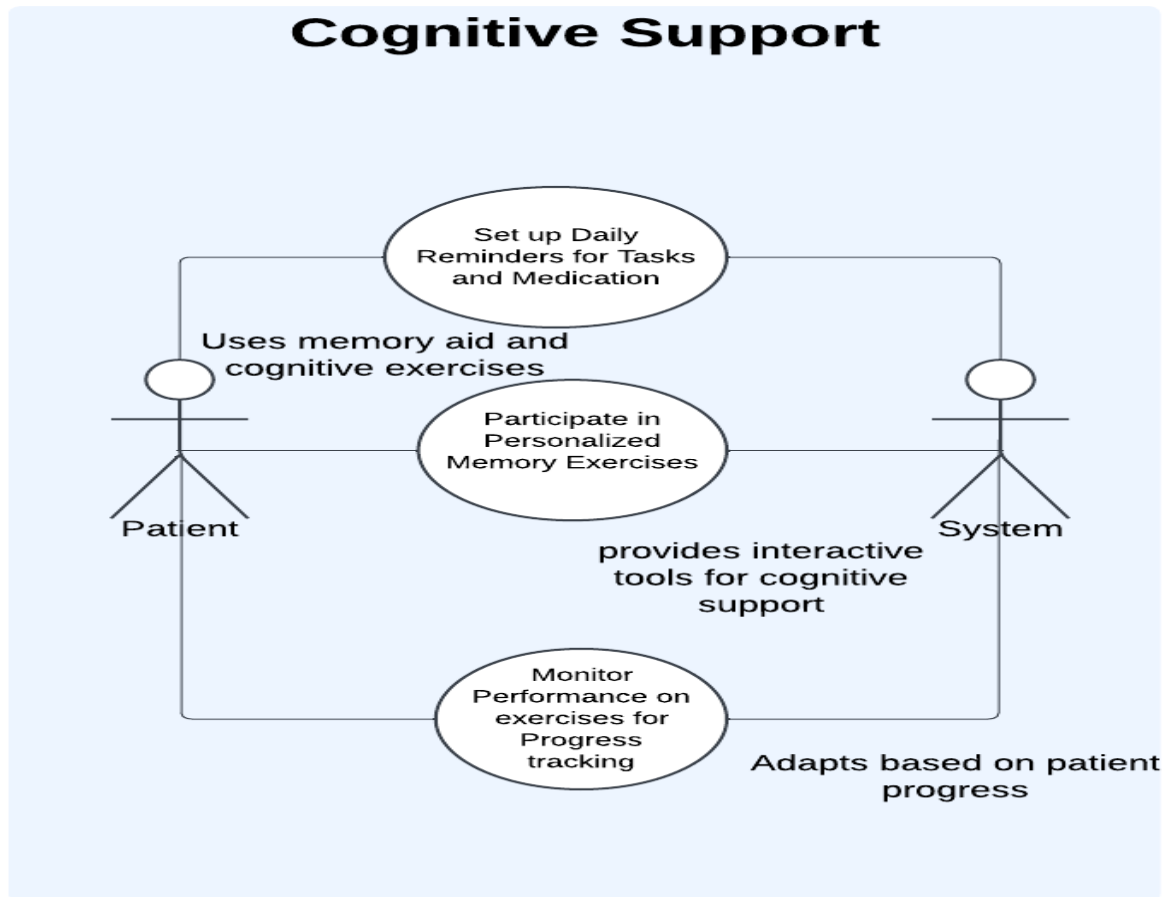
4.1 Early Detection:

- Identify early signs of Alzheimer's through AI models that utilize LangChain for advanced natural language processing and data integration from clinical and imaging datasets.



4.2 Cognitive Support:

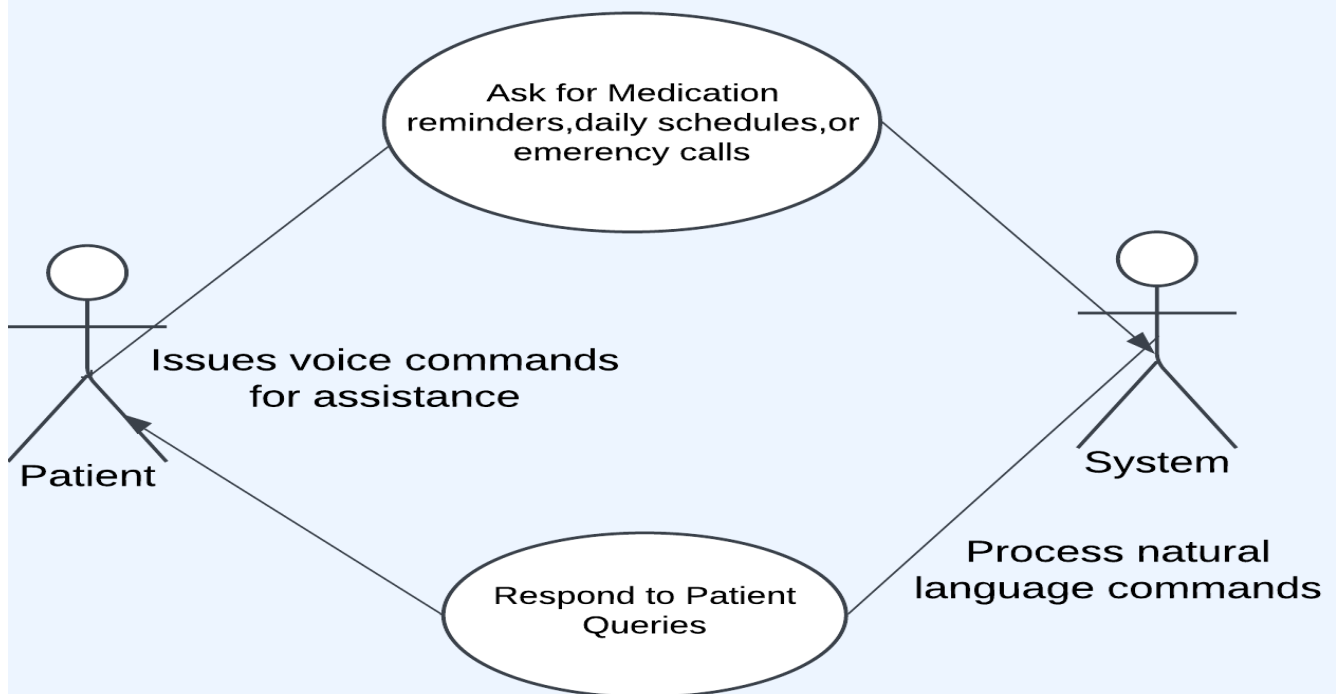
- Offer personalized cognitive exercises and memory aids tailored to the user's cognitive condition using AI-powered LangChain tools.



4.3 Voice Assistance:

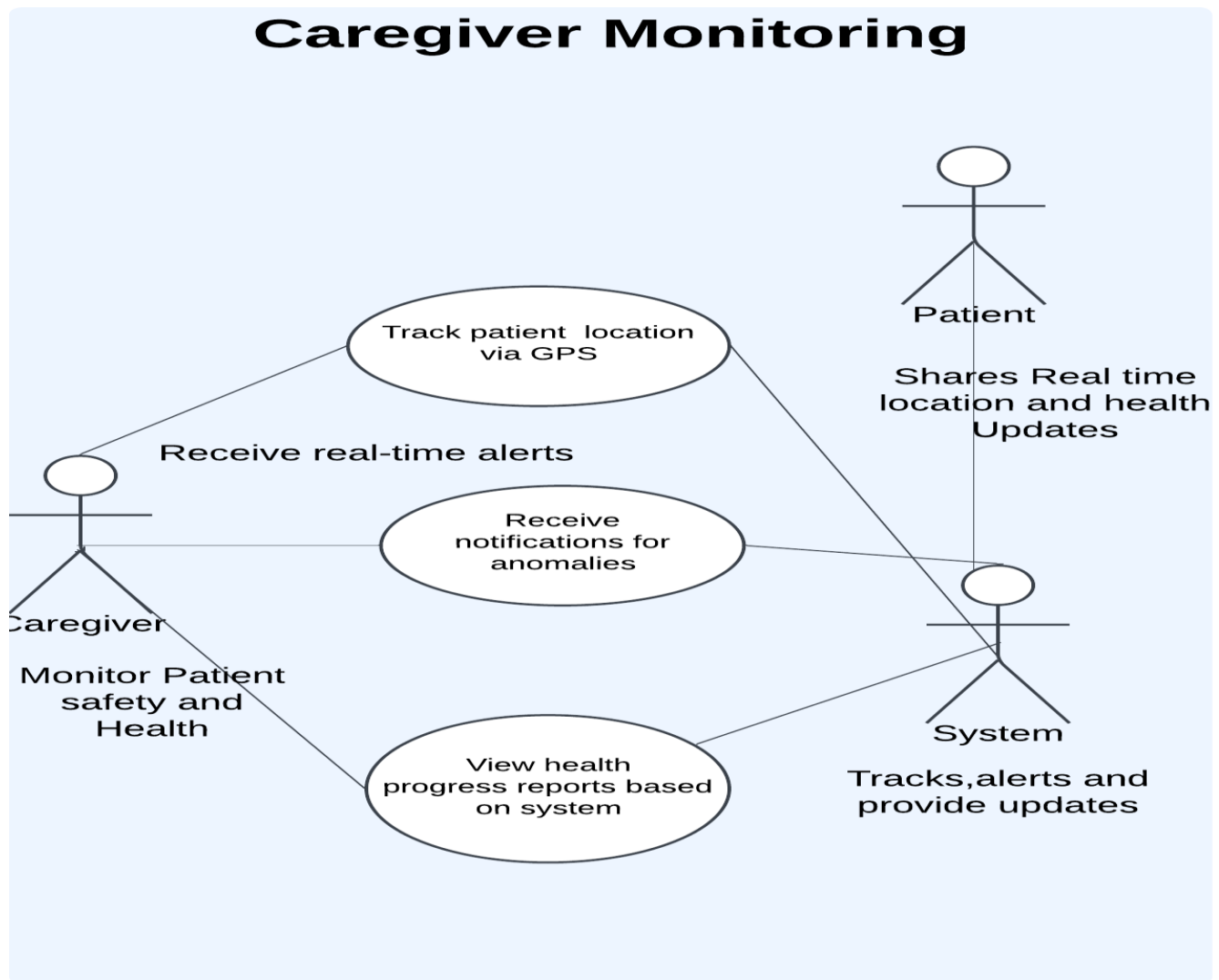
- Integrate a voice assistant powered by LangChain for seamless natural language understanding, providing reminders, task management, and cognitive support.

Voice Assistant



4.4 Caregiver Monitoring:

Allow caregivers to monitor patients via GPS and receive health notifications.



5. Non-functional Requirements

5.1 Performance Specifications

AI models are required to process data and provide results in under 10 seconds.

5.2 Safety Specifications

Sensitive information must be protected through data encryption and multi-layered security measures.

5.3 Security Specifications

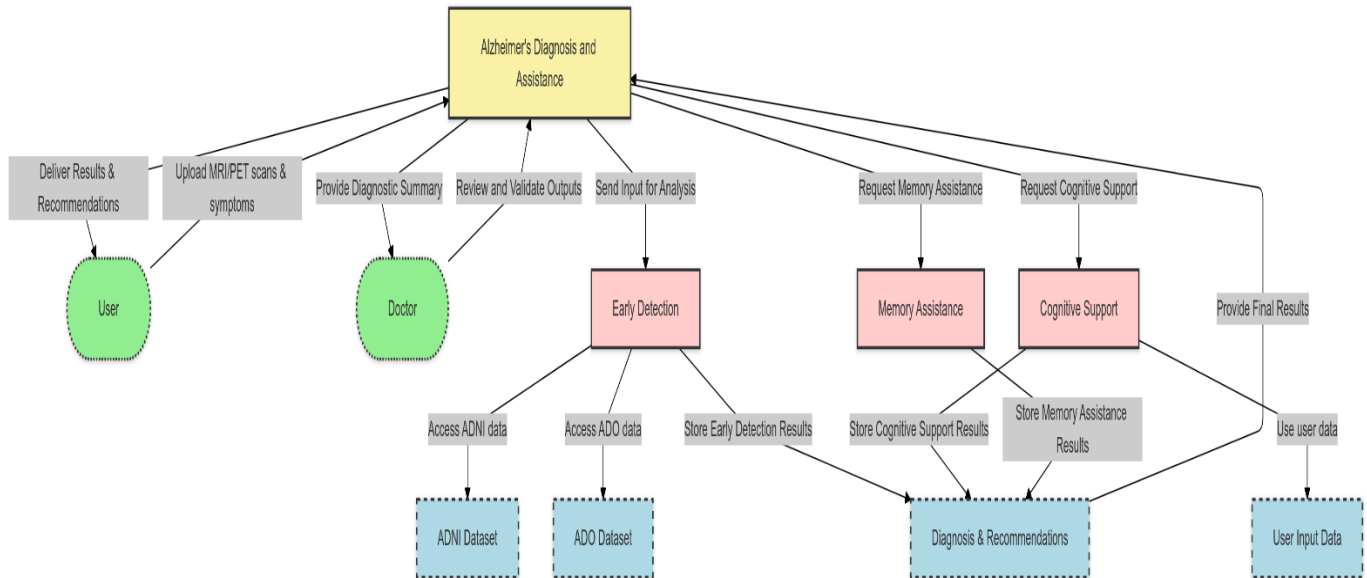
Access to data should be managed through role-based access control.

Regular assessments for vulnerabilities should be conducted to protect the system.

5.4 Usability

The interface should be user-friendly, specifically designed to accommodate elderly patients with visual and cognitive challenges.

Following is the data flow diagram for Alzheimer's Aid System:



6. Assumptions and Dependencies

- Reliable access to datasets (ADNI, KAGGLE) for training AI models.
- Availability of GPS and internet services for tracking functionalities.

7.References

REF1	Project Proposal	OCT.26, 2024	Project Repository
REF2	Alzheimer's Disease Neuroimaging Initiative (ADNI) Dataset	2024	ADNI Website
REF3	Kaggle Alzheimer's Disease Dataset	2024	Kaggle Dataset
REF4	Neo4j Graph Database Documentation	2024	Neo4j Website

8.Appendices

Supporting Details:

8.1 Tools and Technologies

- **LangChain Technology:** Used for processing and managing complex natural language tasks, integrating seamlessly with clinical datasets, and enhancing AI model interpretability.
- **NLP Models:** Employed for the voice assistant to enable natural and interactive conversations.
- **Encryption Methods:** AES-256 and TLS for secure communication and data storage.

8.2 Datasets

- **ADNI Dataset:** Alzheimer's Disease Neuroimaging Initiative provides high-quality MRI, PET scans, and clinical data essential for training and validating AI models.
- **Kaggle Dataset:** Supplemental dataset for additional training and testing purposes, focusing on Alzheimer's progression.

8.3 Security Standards

The system will comply with:

- **HIPAA** (Health Insurance Portability and Accountability Act) for handling sensitive health data.
- **GDPR** (General Data Protection Regulation) for ensuring data privacy, especially if deployed in regions under European jurisdiction.

8.4 Tools and Technologies

- **Graph Algorithms:** Implemented for analyzing patient data relationships (e.g., Graph Traversals, Community Detection).
- **NLP Models:** Used in the voice assistant for natural language understanding and interaction.
- **Encryption Methods:** AES-256 and TLS for secure communication and data storage.

8.5 User Interfaces

- **Patient Interface:** Large, accessible buttons with voice guidance for navigation.
- **Caregiver Dashboard:** Real-time monitoring, alerts, and health data visualization.
- **Healthcare Professional Panel:** Detailed data reports and analysis for clinical use.

8.6 Testing Plans

- **Unit Testing:** For individual modules such as authentication, graph queries, and NLP components.
- **Integration Testing:** Ensure seamless data flow between modules like patient input and caregiver monitoring.
- **User Acceptance Testing (UAT):** Validate usability with feedback from actual patients and caregivers.

8.7 Future Extensions

- **Telemedicine Integration:** For direct consultations with healthcare professionals.
- **Expanded Conditions:** Extend support to other neurodegenerative disorders such as Parkinson's disease.